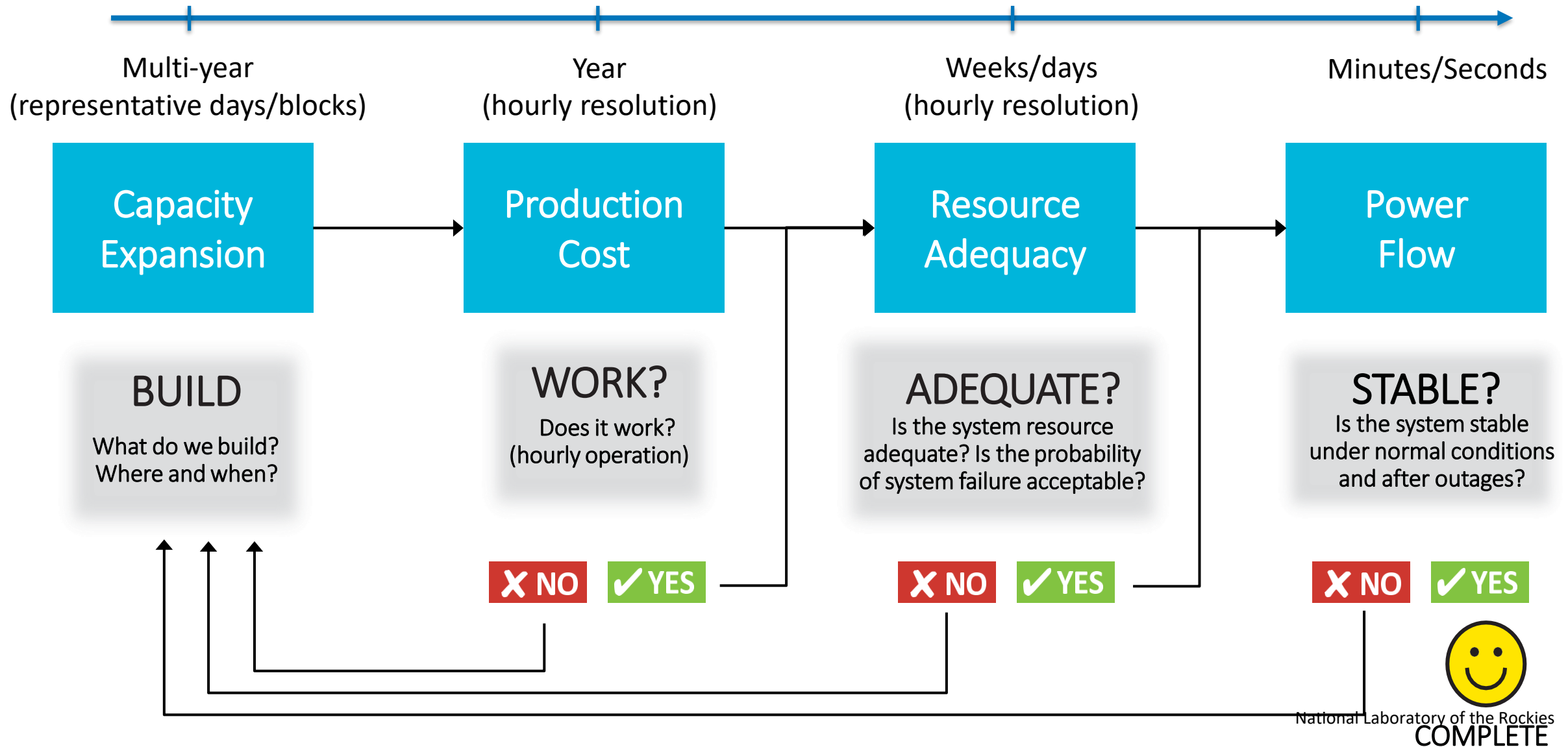


NATIONAL LABORATORY OF THE ROCKIES

West Africa Battery Energy Storage Systems (BESS)
Capacity Building: Energy Storage Analysis –Storage
Deployment Optimization Model (SDOM)

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A Bulk Power Sector Modeling Approach

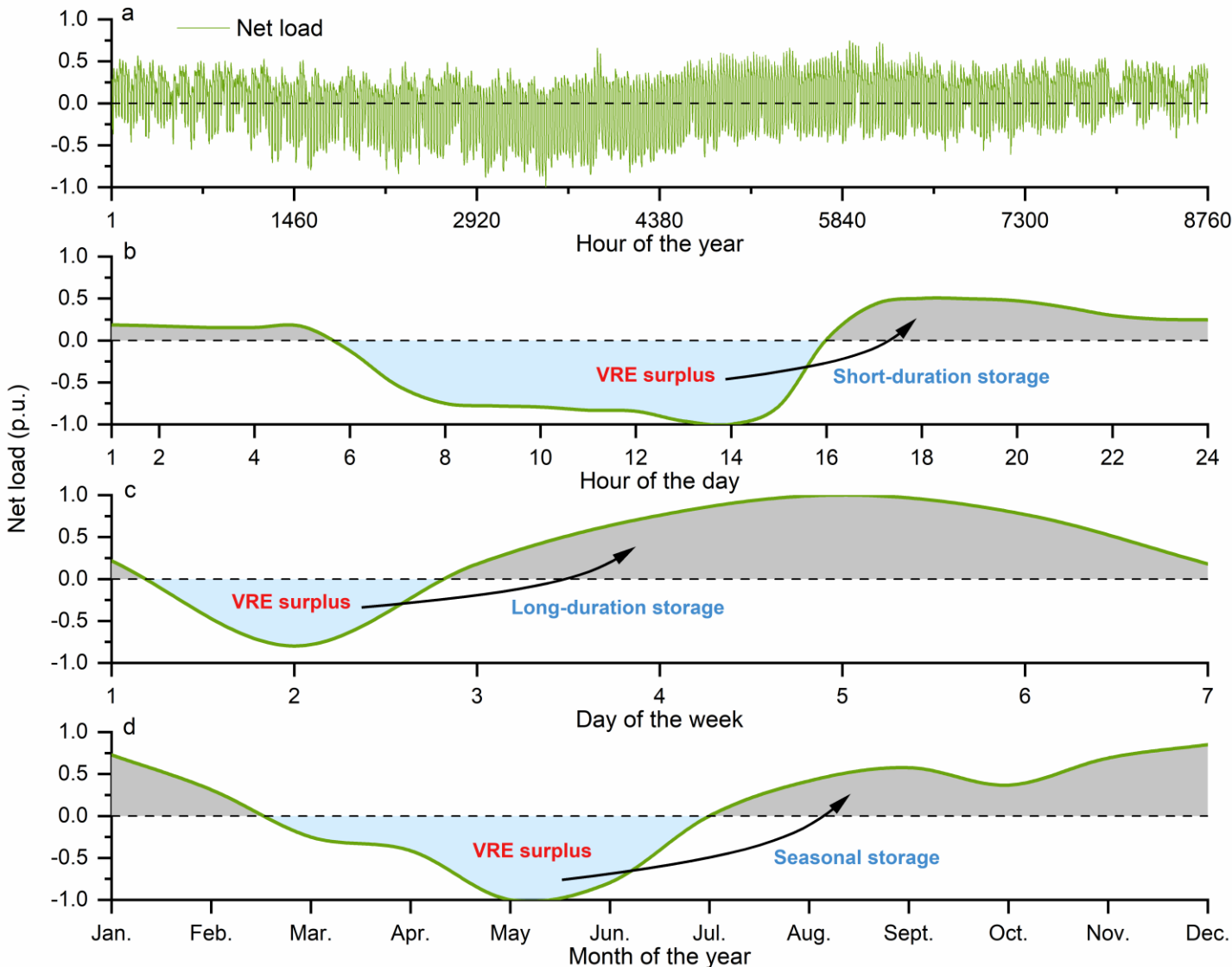


Wind and Solar PV Integration: The Balance Challenge (Economically Balancing Supply and Demand)

Multi-scale energy storage needs for 95% carbon-free CAISO power system

(28.4% wind and 51.5% solar PV energy share)

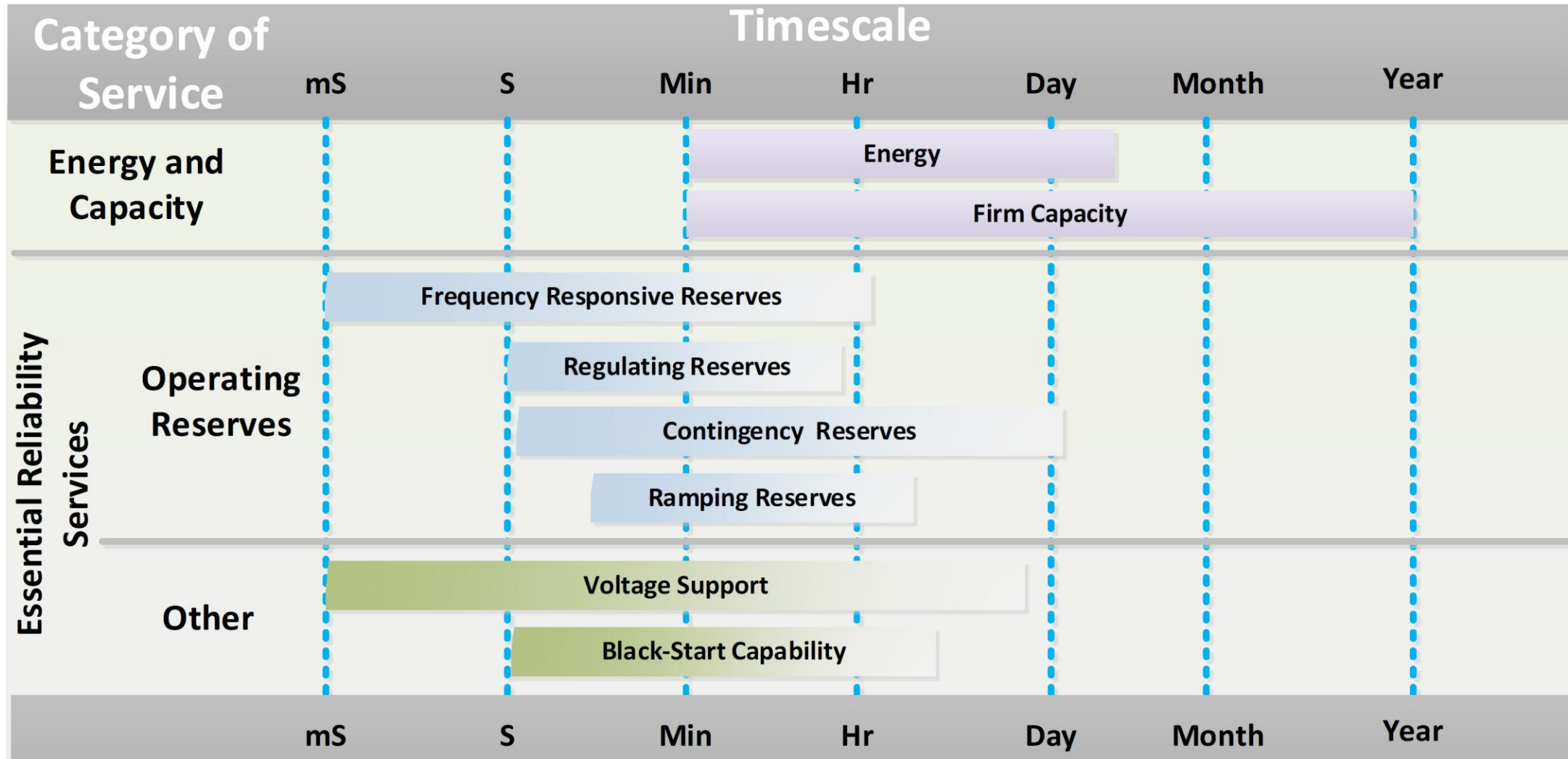
Ref : Guerra, O. J. [Beyond short-duration energy storage](#). Nature Energy 6, 460–461 (2021).



1. The shorter-term fluctuations of VRE resources and the late-afternoon net-load ramps that can occur in systems with significant solar generation
2. The diurnal mismatch that results from the demand peak typically occurring 4 to 6 h after the solar peak and the tendency of wind to generate more at night in many regions of the United States
3. A significant seasonal mismatch among wind, solar, and demand patterns

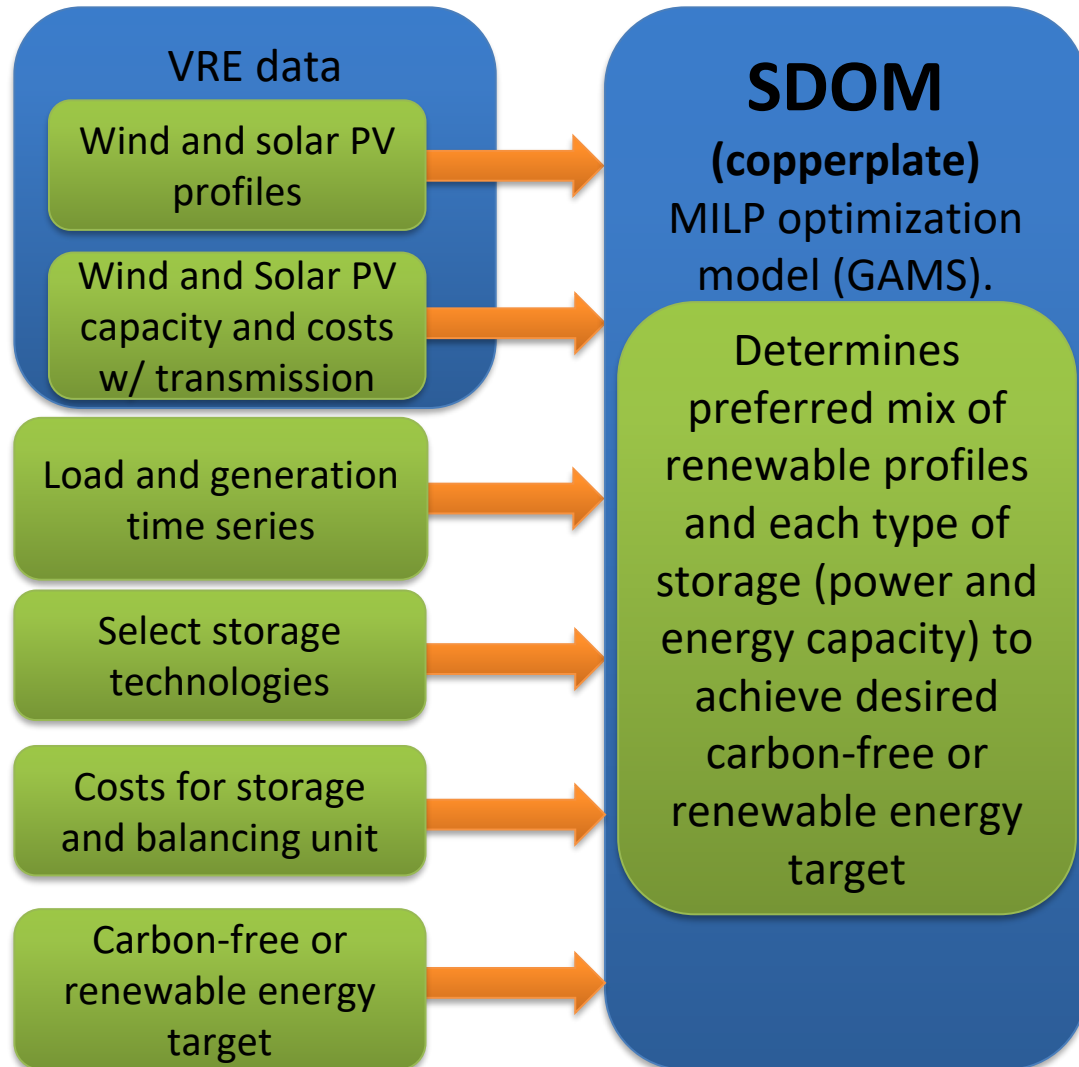
P. Denholm, D.J. Arent, S.F. Baldwin, D.E. Bilello, G.L. Brinkman, J.M. Cochran, et al. [The challenges of achieving a 100% renewable electricity system in the United States](#). Joule, 5 (2021), pp. 1331-1352

Main Storage Services Procured in the U.S. Power System.



Denholm, Paul L., Sun, Yinong, and Mai, Trieu T. [An Introduction to Grid Services: Concepts, Technical Requirements, and Provision from Wind](#). United States: N. p., 2019. Web. doi:10.2172/1493402.

Storage Deployment Optimization Model (SDOM): Modelling Framework



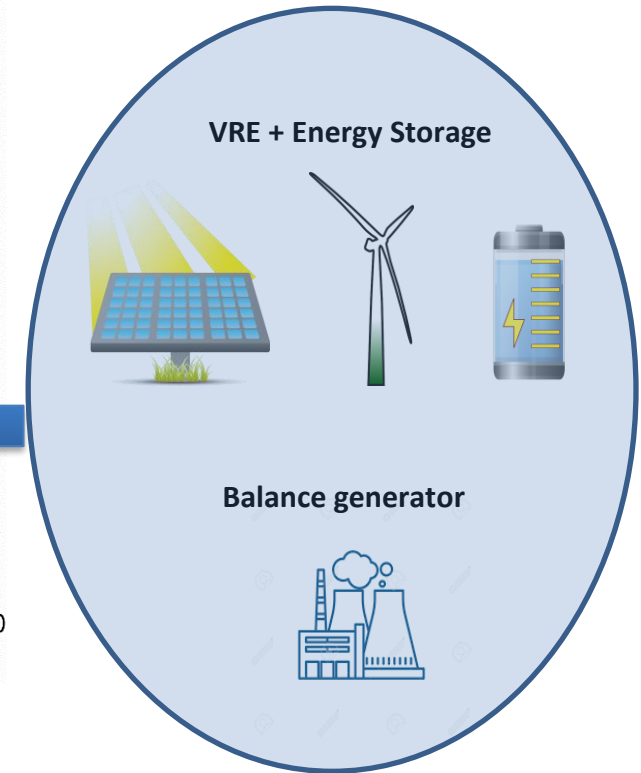
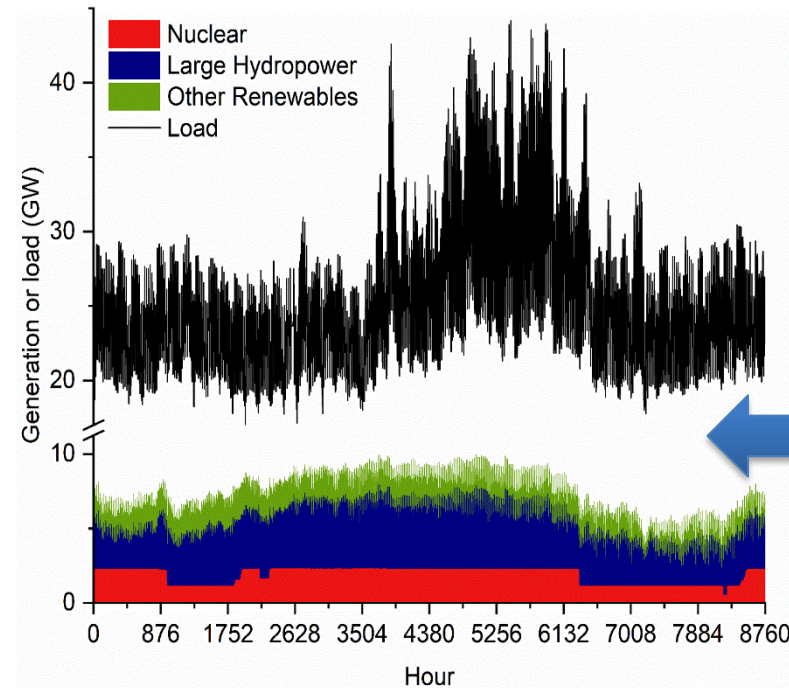
Prior Applications:

- Energy storage requirements of carbon-free and renewable power systems in United States, including Li-Ion, CAES, PHS, and Hydrogen storage technologies. Preliminary results are available for CAISO and ERCOT but the study will be extended to SPP, PJM, MISO, NYISO, and ISO-NE
- Research questions:
 - How should a power system be built to minimize total system cost?
 - How much wind and solar PV generation capacity is required to achieve a given RPS target?
 - How much storage power and energy capacity is required to achieve a given RPS target?
- Client and stakeholder involvement: DOE

VRE: variable renewable energy, RPS: renewable portfolio standards, CAES: compressed-air energy storage, PHS: Pumped hydro storage

What does **SDOM**TM do?

- The **Storage Deployment Optimization Model (SDOM)** is a flexible, open-source platform created by NREL that can evaluate how different storage technologies perform across a range of integration scenarios.
- It works by **identifying the least-cost mix of storage and generation technologies** that meet demand while respecting system constraints and a given target of renewable generation.
- It can be used in long-term planning exercises to explore how much storage a system may need, what type, and how it would be used.
- All the results presented in this analysis were generated using SDOM.



Learn more:

- [SDOM GitHub repository](#)
- [DOE CODE entry \(OSTI\)](#)

Example SDOM Analysis: RELAC

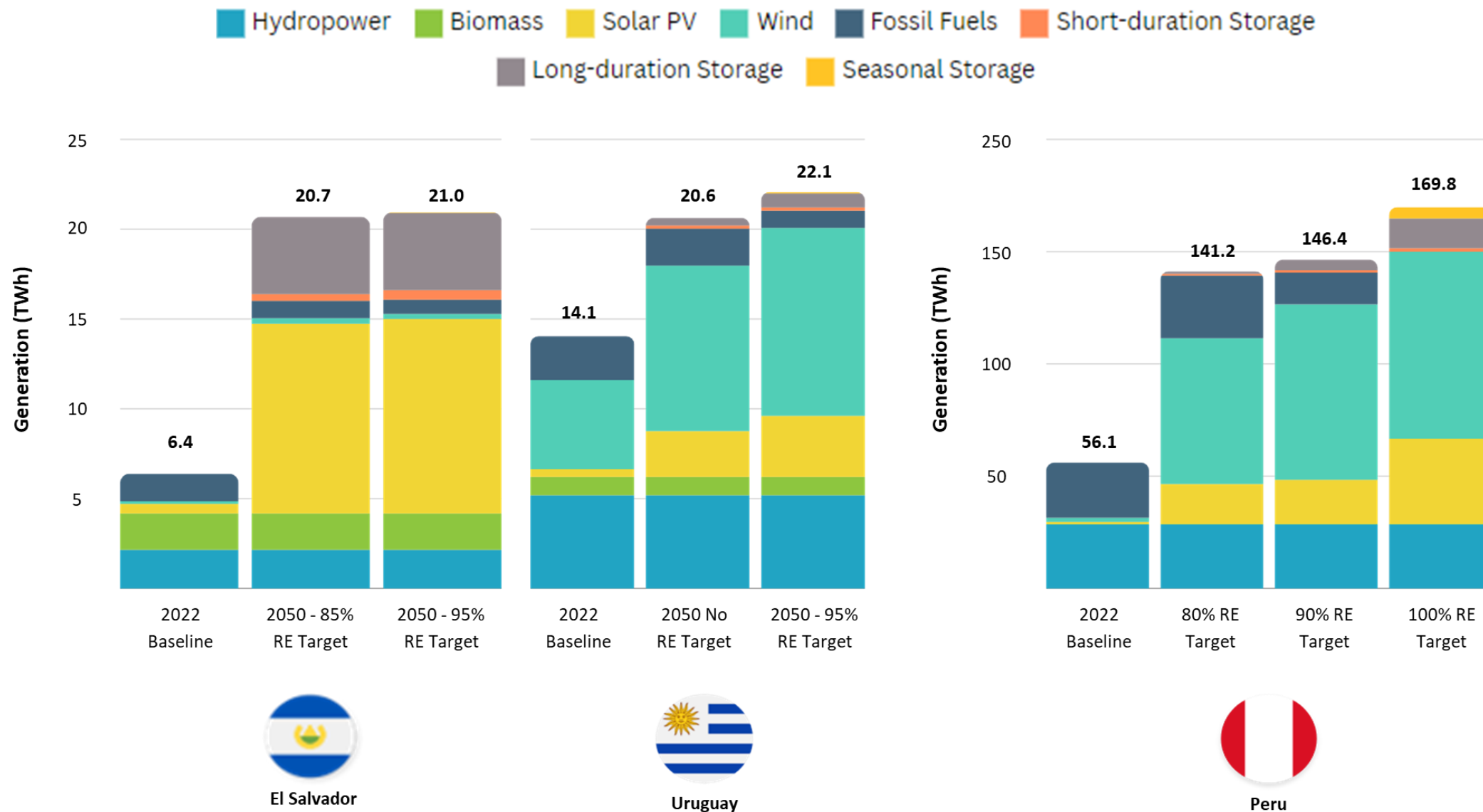


Image by Andrew Bilich (NREL)

Renewable Energy in Latin America and the Caribbean (RELAC)

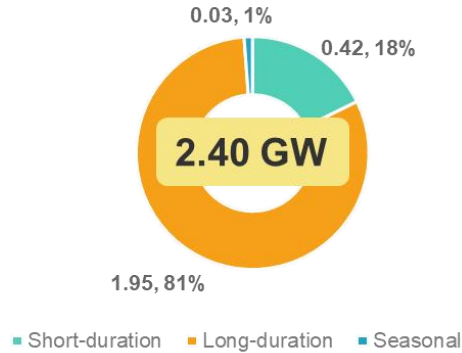
NREL developed initial modeling for **Uruguay, Peru, and El Salvador** to assess different scenarios for energy storage (including short, long, and seasonal duration) that support renewables integration, reduce curtailment, and increase grid stability through 2050.

Example SDOM Analysis: RELAC

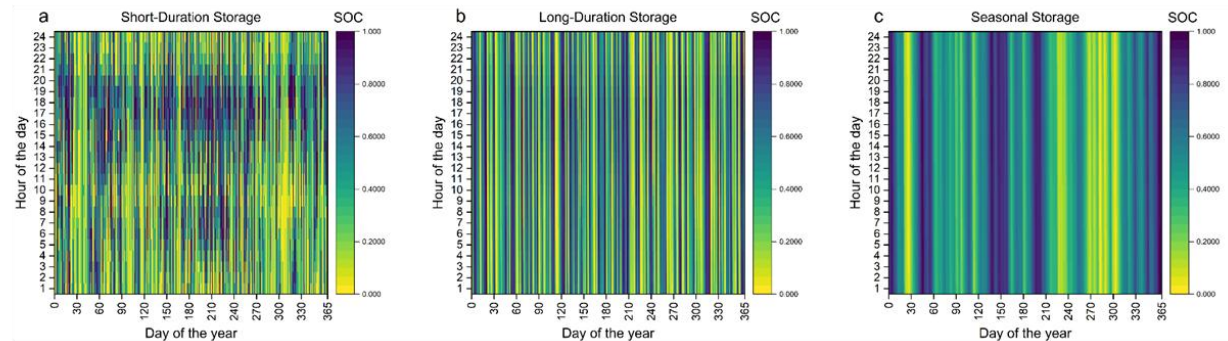
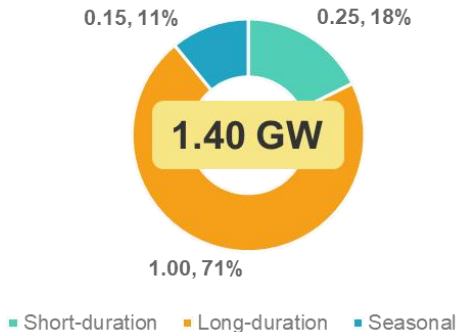
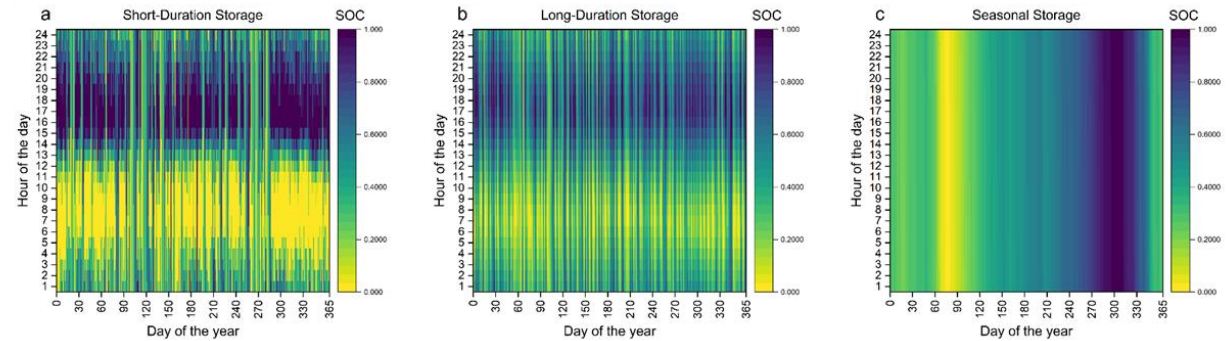
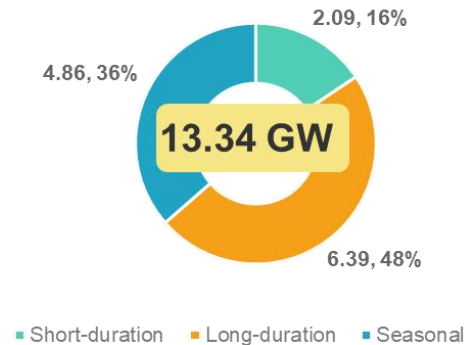
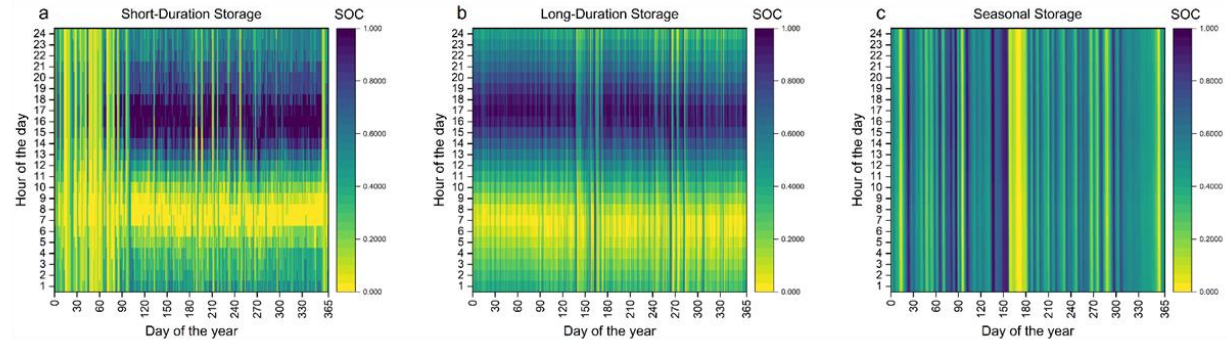


Example SDOM Analysis: RELAC

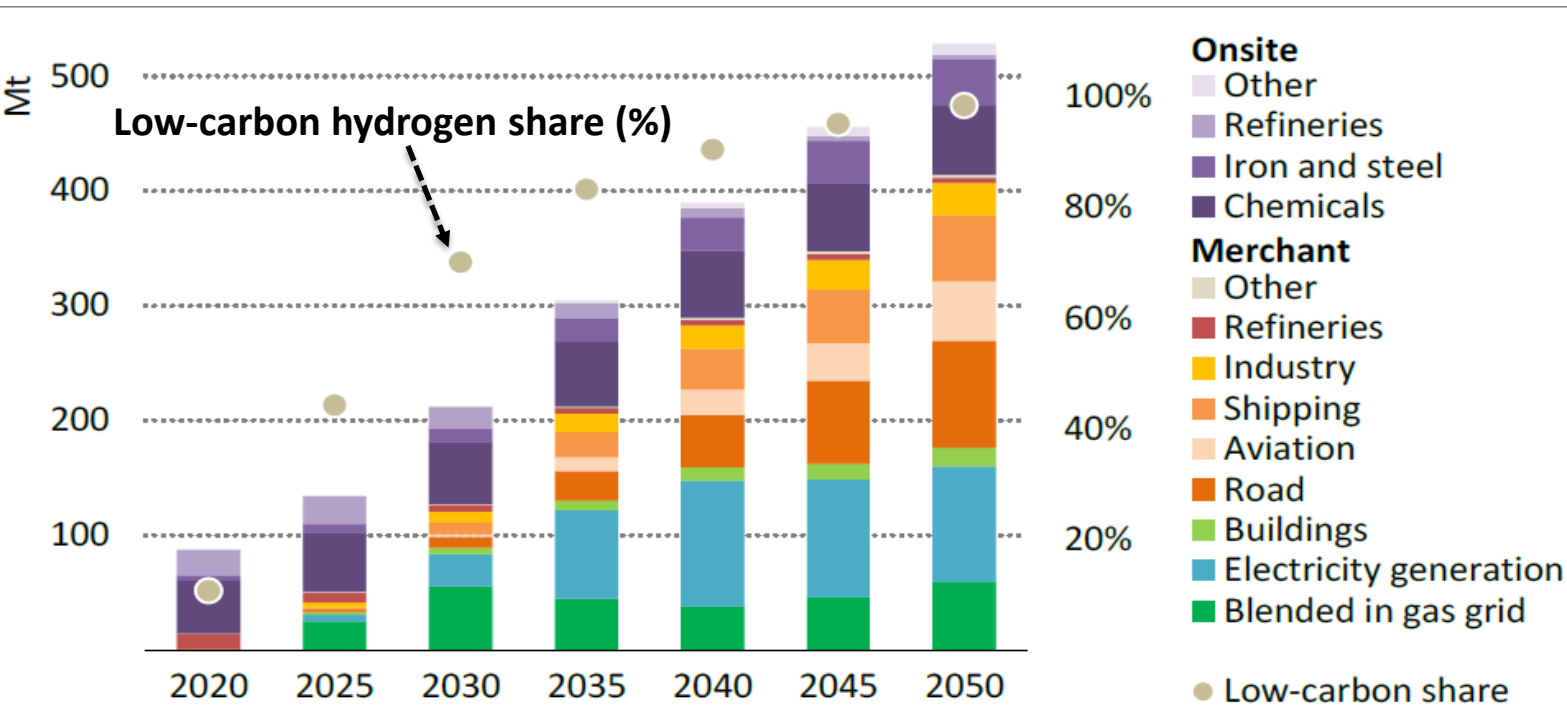
2050 - Total Storage Capacity (GW)



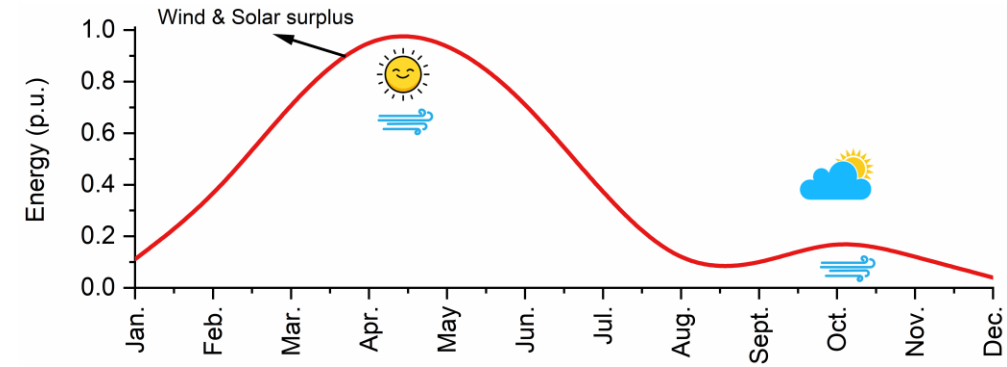
Annual Hourly Storage State of Charge by Storage Type
(yellow = 0% , dark blue = 100%)



Seasonal Storage: Hydrogen as an Enabling Technology for the Integration of VRE



Opportunities for seasonal energy storage



Grid-integrated seasonal energy storage can reshape seasonal fluctuations of variable and uncertain power generation by reducing energy curtailment, replacing peak generation capacity, and providing transmission benefits.

Ref: <https://www.iea.org/reports/net-zero-by-2050>

O.J. Guerra, J. Zhang, J. Eichman, P. Denholm, J. Kurtz, B.-M. Hodge. The Value of Seasonal Energy Storage Technologies for the Integration of Wind and Solar Power. Energy & Environmental Science (2020), <https://pubs.rsc.org/en/content/articlelanding/2020/EE/D0EE00771D#!divCitation>

Q&A



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Thank you

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